Invertix: Data-Center Siting & Power

AI is driving a wave of new data centers, and the binding constraint is increasingly electricity. Siting one is a multi-waytrade-off between price, carbon, grid congestion and connectivity. Can you help someone reason about where to build and how to power it?

WORTH EXPLORING

- Recommend locations for a given data-center size and explain the trade-offs.
- Plan a supply mix of grid, PPA and on-site generation against cost and carbon.
- Overlay capacity, prices, carbon and congestion to reveal good vs bad sites.

PyPSA-Eur   Ember data   OpenStreetMap   IEA Energy & AI   Google Alphaearth

Invertix Open Track:
Satellite Data for Solar

Satellites capture irradiance, cloud cover, and atmospheric conditions across the globe — data that ground sensors can't match in coverage or scale.
Can you turn it into something operators actually use?

WORTH EXPLORING

- A generation forecast that beats persistence using satellite-derived irradiance.
- A nowcast for intraday trading or grid balancing decisions. reveal good vs bad sites.
- Any tool that takes raw satellite feeds and turns them into an operational output.

Copernicus / Sentinel   NASA POWER   EUMETSAT   Google Earth Engine

ENERPARC: Digital Twins of Solar Plants

Find hidden underperformance in 10 year real solar plant monitoring data using ML

Solar plants are historically mostly judged by the Performance Ratio (PR). There is a clear need for advanced, data-driven analytics on inverter level that go beyond PR. Can you make model-based approaches analyzing degradation trends and find out the financial impacts from quality issues, downtimes and grid curtailments — and prove both with numbers?

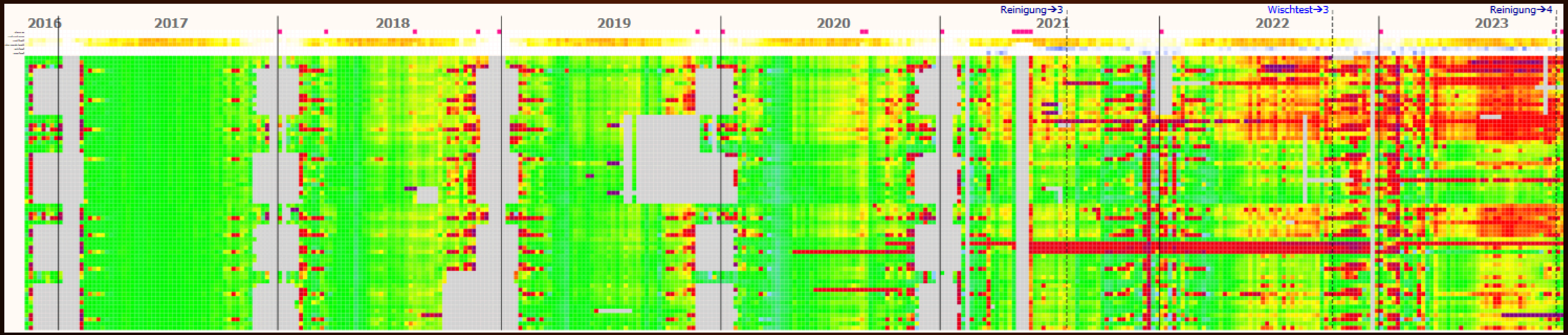
WORTH EXPLORING

- Build a per-inverter ML power model (optional also for current and voltage)
- Benchmark individual inverter models (trained with 1st year operation data) against real inverter dataset of 10 years! Compare results for different module types over the whole timeline.

`pvlib-python` `scikit-learn` `or LightGBM` `Embedding ticket and errorcode info`

ENERPARC: Digital Twins of Solar Plants

Long Term Performance Tracking:



ENERPARC Open Track:

BUILT USEFULL AGENTS FOR THE PROVIDED DATASET

Build your own agents on top of 10 years of real solar plant monitoring and operational data from two utility-scale PV plants. Connect minute-resolution inverter telemetry, error codes, service tickets and plant design info — and turn it into something an O&M team would actually use.

WORTH EXPLORING

- → Inverter failure detection
- → Linking errorcode events to impact on inverter production
- → Chatbot giving answers & visualisations of monitoring data
- → Any agent that takes raw plant data and turns it into an O&M decision.

EC 61724 PVsyst pvlib IEA PVPS Task 13 DuckDB LangGraph

e·on: Grid Operation Agents

A grid must survive losing any single line or generator without anything else

overloading : staying "N-1 secure." When something trips, an operator has minutes to switch lines, redispatch generators or curtail output. Can an agent reason through that, propose safe low-cost actions, and explain them?

WORTH EXPLORING

→ Take an overloaded grid back to a safe state, narrating its reasoning.

→ Screen thousands of "what if this fails" cases; solve only the dangerous ones.

→ Benchmark an LLM agent against a rule-based or optimization baseline.

pandapower   Grid2Op / L2RPN   PyPSA-Eur   real-time decisions

e·on Open Track: Grid Foundation Models

A pretrained grid model can predict powerflow dispatch far faster than a solver.
The interesting question is what near-instant prediction unlocks — interactive what-ifs, rapid screening, smarter optimizers. Can you build something useful on top of one, and stress-test where it holds up?

WORTH EXPLORING

- A “drag a load, see the grid respond” explorer powered by a fast model. Rank risky failure cases, then verify
- the worst with a real solver.
- An honest evaluation: how accurate is it, and where does it break?
- Anything that wouldn't be possible if inference took 10 seconds instead of 10 milliseconds.

`GridSFM_Open` `pandapower` `gridfm-graphkit` `OPFData`